

A proposed counterexample to Proposition 2.11

Classicism, 1 July 2022 draft, p. 30 and footnote 42

The following construction appears to refute Proposition 2.11, not just the normalization in footnote 42. It satisfies Classicism and, at every world, BF, Boolean Completeness, and Atomicity. Nevertheless, Rigid Comprehension fails for properties of propositions.

The construction uses a nonprincipal ultrafilter, but only ultralimits in fixed finite sets. We give the carriers and action maps, verify closure and the hypotheses, and exhibit the failed comprehension instance.

1. The typed model. Work in ZFC with a nonprincipal ultrafilter $\mathcal{U}$ on the positive integers. Take one full extensional simple-type model M , with $M_e = \{*\}$, $M_t = \{0, 1\}$, and full function spaces. Each fixed type carrier is finite. For a sequence (a_n) in a finite nonempty set A , define $\lim_{\mathcal{U}} a_n = a$ when $\{n : a_n = a\} \in \mathcal{U}$. Exactly one cell of a finite partition belongs to an ultrafilter, so this value exists and is unique. If $f_n : A \rightarrow B$ and $a_n \in A$, with A, B finite, then

$$(1) \quad \lim_{\mathcal{U}} f_n(a_n) = (\lim_{\mathcal{U}} f_n)(\lim_{\mathcal{U}} a_n).$$

Intersect the two $\mathcal{U}$ -large sets on which f_n and a_n take their limiting values to obtain this identity. The same argument applies to each operation on fixed finite sets. Changing finitely many terms does not change a limit, since $\mathcal{U}$ contains every cofinite set.

Use worlds and arrows

$$o \rightarrow s \rightarrow u, \quad o \rightarrow n \quad (n \geq 1),$$

including identities and the composite $o \rightarrow u$. Put M at every n and at u . At s , take $D_s(e) = \{*\}$ and $D_s(t) = \{0, 1\}^2$, with $j_e = \text{id}$ and $j_t(b_s, b_u) = b_u$. Recursively, $D_s(\alpha \rightarrow \beta)$ contains all pairs (g_s, g_u) with $g_s : D_s(\alpha) \rightarrow D_s(\beta)$, $g_u \in M_{\alpha \rightarrow \beta}$, and $j_\beta g_s = g_u j_\alpha$. Define $\text{app}_s((g_s, g_u), x) = g_s(x)$; the action $j_{\alpha \rightarrow \beta}$ returns g_u .

At o , take $D_o(e) = \{*\}$ and

$$D_o(t) = \{0, 1\} \times \{0, 1\} \times \{0, 1\}^{\mathbb{N}_{>0}}, \quad \pi_s(b_o, b_s, (b_n)) = (b_s, \lim_{\mathcal{U}} b_n), \quad \pi_n(b) = b_n.$$

A proposition is true at its current world exactly when its current bit is 1. A root object of type $\alpha \rightarrow \beta$ is a triple $(f_o, F_s, (f_n))$, where

$$f_o : D_o(\alpha) \rightarrow D_o(\beta), \quad F_s \in D_s(\alpha \rightarrow \beta), \quad f_n \in M_{\alpha \rightarrow \beta}.$$

For every $x \in D_o(\alpha)$ it satisfies

$$\pi_s(f_o(x)) = \text{app}_s(F_s, \pi_s x), \quad \pi_n(f_o(x)) = f_n(\pi_n x), \quad j(F_s) = \lim_u f_n.$$

Application uses the current function component. The maps to s and n return the corresponding future components; the map to u is $j\pi_s$.

2. Surjectivity and application. We prove by induction on types that every carrier is nonempty, each $D_s(\alpha)$ is finite, j_α is onto, and

$$x \mapsto (\pi_s x, (\pi_n x)_n)$$

maps $D_o(\alpha)$ onto

$$E_\alpha = \{(a_s, (a_n)) : a_s \in D_s(\alpha), a_n \in M_\alpha, j_\alpha a_s = \lim_u a_n\}.$$

For e these claims follow from the singleton definitions. For t , a compatible pair consists of $a_s = (b_s, b_u)$ and a sequence (b_n) with $b_u = \lim_u b_n$; the root proposition $(0, b_s, (b_n))$ lifts it. The map j_t is onto.

Suppose the assertions hold for α, β . The intermediate carrier at $\alpha \rightarrow \beta$ is a subset of a product of finite function spaces. Given $g_u \in M_{\alpha \rightarrow \beta}$, choose, for each $x \in D_s(\alpha)$, a preimage under j_β of $g_u(j_\alpha x)$. These choices define g_s , so (g_s, g_u) belongs to $D_s(\alpha \rightarrow \beta)$ and maps to g_u .

For root joint surjectivity, let $(F_s, (f_n)) \in E_{\alpha \rightarrow \beta}$. For each $x \in D_o(\alpha)$ the desired outputs are

$$y_s = \text{app}_s(F_s, \pi_s x), \quad y_n = f_n(\pi_n x).$$

They are compatible, since

$$j_\beta y_s = j_{\alpha \rightarrow \beta}(F_s)(j_\alpha \pi_s x) = (\lim_u f_n)(\lim_u \pi_n x) = \lim_u y_n.$$

The induction hypothesis at β supplies a root preimage of this tuple. Choose one for each x ; these choices give the required f_o . This uses ordinary metatheoretic choice in ZFC. Nonemptiness follows at the same step by taking constant functions between inhabited carriers.

Every individual root action is onto. To lift a_s , take the constant sequence $a_n = j_\alpha a_s$ and use joint surjectivity. To prescribe a value at one terminal world n , start with such a constant sequence and change its n th entry. The limit is unchanged. The action to u is onto because it is the composite $j_\alpha \pi_s$. Application commutes with every action by the defining

equations.

The current application function determines each function object. At s , surjectivity of j_α determines g_u from g_s . At o , surjectivity of π_s and each π_n determines F_s and f_n from f_o , using the intermediate result for F_s . Equality here is equality of whole objects, not mere agreement of their current truth extensions.

3. Logical operations and abstraction. For an arrow $h : v \rightarrow w$, write h_α for its action at type α . The truth bit at w of equality at v is

$$[h_\alpha a = h_\alpha b],$$

where brackets denote 1 if the statement is true and 0 otherwise. Negation, conjunction and disjunction act on the truth bits at each world. For $F \in D_v(\alpha \rightarrow t)$, the truth bit at w of $\forall_\alpha F$ is

$$[\text{app}_w(h_{\alpha \rightarrow t} F, a) \text{ is true at } w \text{ for every } a \in D_w(\alpha)].$$

Existential quantification uses *some* in place of *every*.

These prescriptions define admissible objects. At s , their future components are the corresponding operations in M . At the root, the required u -bit is the ultralimit of the terminal bits. For equality this follows from the finite-set limit identity applied to the equality test. For quantification, the terminal predicate functions range over the finite set $M_{\alpha \rightarrow t}$; on a $\mathcal{U}$ -large set the entire function equals its limit. Universal and existential quantification over the fixed finite M_α therefore give the same limiting bit. The Boolean cases follow from (1). Their curried function components at future worlds are specified by the same prescriptions, so application naturality and the higher-type compatibility equations hold as well.

Interpret a variable by its assigned object, application by the current application function, and a constant at a future world by transporting its root value. At v , interpret $\lambda x.A$ by its current function

$$a \mapsto [[A]]_v^{g[x \mapsto a]}$$

together with the corresponding future functions. Simultaneous induction on terms proves membership in the prescribed carriers and

$$(2) \quad h_\alpha [[A]]_v^g = [[A]]_w^{hg} \quad (A : \alpha, h : v \rightarrow w).$$

For abstraction, apply the induction hypothesis for the body to $g[x \mapsto a]$; its transported assignment is $(hg)[x \mapsto h_\sigma a]$, where $x : \sigma$. This is the compatibility equation for the current and future functions. Application uses its defining naturality equation, and logical constants use the preceding paragraph.

At the root one must also check the ultralimit condition between the terminal functions and the u -function. Each term involves only finitely many free variables and nonlogical constants. Intersect the $\mathcal{U}$ -large sets on which their terminal values equal their u -values. On this intersection the interpretations in the identical full model M agree. This proves the limit condition for the whole denotation, including function-valued terms. Thus the abstraction functions above belong to the required carriers.

For comparison with Definitions 3.18–3.20, encode $p \in D_v(t)$ as the set of outgoing arrows $h : v \rightarrow w$ for which $h_t p$ is true at w . Recursively encode a function F by the graph

$$(h, a) \mapsto \text{app}_w(h_{\alpha \rightarrow \beta} F, a) \quad (a \in D_w(\alpha)).$$

Arguments and outputs are encoded at their respective types. The proposition encoding is injective: it records every free bit. The function encoding is injective by induction, since it includes current application and all future application. Transport becomes the paper's inverse-image action on propositions and precomposition action on graphs. The truth clauses above are its six primitive clauses, and (2) supplies its abstraction clause. We therefore obtain an action model in the paper's sense. Theorem 3.23 gives Classicism.

In this model $\Box p$, defined as $p = \top$, is true at v just when p is true at every world accessible from v : equality with $\top$ compares the whole outgoing truth profile.

4. BF, Boolean Completeness, and Atomicity. For BF, suppose $\forall x \Box P(x)$ holds at v . Given $h : v \rightarrow w$ and $a \in D_w(\alpha)$, choose $b \in D_v(\alpha)$ with $h_\alpha b = a$. The assumption at b gives $P(a)$ at w , with the other parameters transported along h . Since a and h were arbitrary, $\Box \forall x P(x)$ holds at v . This is the argument in Proposition 3.24(ii); it works at every world and type.

For a relational type $\tau = \sigma_1 \rightarrow \cdots \rightarrow \sigma_k \rightarrow t$, put $T_v = \prod_i D_v(\sigma_i)$. Simultaneous induction on the curried result type gives

$$M_\tau \cong 2^{T_u}, \quad D_s(\tau) \cong 2^{T_s} \times 2^{T_u},$$

$$D_o(\tau) \cong 2^{T_o} \times \{(S, (N_n)) : S \in D_s(\tau), N_n \in M_\tau, jS = \lim_u N_n\}.$$

The case $k = 0$ is the specified proposition carrier. At the arrow step, choose a current truth relation and compatible future operators. For each current argument, the result-type induction supplies the unique output with that truth section and those future components; ultralimit commutation with application gives compatibility. Conversely, application recovers these data uniquely. This also shows that the current relation at s is independent of its terminal operator.

All T_n equal the fixed finite T_u . Thus (N_n) determines the terminal component of S , leaving its current relation free, and

$$D_o(\tau) \cong 2^{T_o} \times 2^{T_s} \times \prod_{n \geq 1} 2^{T_n}.$$

The Boolean operations act coordinatewise. The source order $A \leq B$, meaning $B = A \vee B$ as whole objects, is therefore the product order. The root algebra is complete and atomic; at the other worlds the corresponding algebras are finite. Each factor is the powerset algebra of its tuple set. Arbitrary meets and joins are intersections and unions in each free coordinate. Each nonzero element contains a tuple in one of those coordinates; the element supported on that one tuple in that one coordinate is an atom below it. The empty tuple is used when $k = 0$, so this includes propositions.

To check the source formulas, let $Q \in D_v(\tau \rightarrow t)$ and form $S_Q = \{a \in D_v(\tau) : Qa \text{ is true at } v\}$. Its internal meet m belongs to $D_v(\tau)$. For each z in that carrier,

$$(\text{for every } a \in S_Q, z \leq a) \iff z \leq m.$$

This is the $\text{GLB}_\tau(m, Q)$ clause, including when S_Q is empty. The source formula

$$\text{Atom}_\tau(y) \equiv \forall z((z \leq y \ \& \ z \neq y) \leftrightarrow z \leq \sim_\tau z)$$

says that the strict lower elements of y are precisely bottom. It excludes $y = \perp$ and holds exactly at the ordinary nonzero atoms. Hence Boolean Completeness and Atomicity hold at every world and every relational type, as do their boxed forms and $\Box\text{BF}$. A GLB is chosen separately at the world where its existence is asserted. Neither Boolean Completeness nor its necessitation requires the transport of that GLB to be the GLB of the transported family.

5. The normalization loses its witness. Put $w = (0, 1, (0)_n)$ and $q = (0, 0, (1)_n)$. The proposition w is a root Boolean atom selecting s (not a true root-world proposition). Since $\pi_s(q) = (0, 1)$ is an atom of $\{0, 1\}^2$, $w \leq \text{Atom}(q)$. But the actual extension of $F(p) := (w \leq (p = q))$ is

$$\{p : p_s = 0 \text{ and } \{n : p_n = 1\} \in \mathcal{U}\}.$$

Its meet is $\perp$: the root bit can be zero, the s -bit is always zero, and each fixed n -bit can be zero in a member, using a cofinite set omitting n . Every member maps at s to $(0, 1)$, whereas its meet maps to $(0, 0)$. Thus $w \not\leq (q = \bigwedge F)$, and the replacement is not possibly atomic. Completeness of both algebras does not make the surjective Boolean homomorphism π_s preserve infinite meets. Indeed, the ambient intersection of the equality fiber, as sets of

outgoing arrows, is $\{o \rightarrow u\}$. This is not a root proposition: zero terminal bits force a zero u -bit. Its greatest available lower bound is empty. The internal meet is not the ambient intersection.

6. Rigid Comprehension itself fails. For a unary property the source definitions are

$$\begin{aligned}\text{Persistent}(Z) &\equiv Z \leq (\lambda x. \Box Zx), \\ \text{Inextensible}(Z) &\equiv \Box \forall Y (\forall x (Zx \rightarrow \Box Yx) \rightarrow Z \leq Y).\end{aligned}$$

A rigid property satisfies both. Persistence says that at every accessible world, each instance remains an instance along every further arrow. The order $Z \leq Y$ at v means inclusion of their extensions at every world accessible from v , not just at v . Write Z_v for the current extension of the transported property.

Write $a_0 = (0, 0) \in D_s(t)$ and set $A_0 = \pi_s^{-1}(\{a_0\})$. Its s -image is $\{a_0\}$ and its u -image is $\{0\}$, but each n -image is all of $\{0, 1\}$: a sequence supported at a single n still has ultra-limit zero. Choose $X : t \rightarrow t$ with root extension A_0 and all future extensions empty. The displayed Boolean decomposition admits this property.

Suppose that a rigid property Z is coextensive with X at o . Persistence forces $Z_n = \{0, 1\}$ for every n and $a_0 \in Z_s$. Carrier compatibility then forces $Z_u = \{0, 1\}$: its extension is the ultralimit of the Z_n , not the image along an arrow from a leaf to u . Now choose $B : t \rightarrow t$ with extensions

$$B_o = A_0, \quad B_s = \{a_0\}, \quad B_n = B_u = \{0, 1\}.$$

Every actual Z -instance necessarily satisfies B . Root inextensibility gives $Z \leq B$, hence $Z_s \subseteq \{a_0\}$; persistence gives the reverse inclusion. At s , however, the haecceity Y of a_0 has extensions $Y_s = \{a_0\}$ and $Y_u = \{0\}$. Every s -instance of Z necessarily satisfies Y , but $Z \leq Y$ fails at s : the element $1 \in Z_u$ is not in Y_u . This contradicts the s -instance of the *outer-boxed* inextensibility condition. No rigid property is root-coextensive with X . The individual sort can remain a singleton throughout. The outer box is essential: B satisfies persistence and the unboxed inextensibility clause at o , but fails source-defined inextensibility because that clause must also hold at s .

7. The full setup of footnote 42 is realized. For $a \in A_0$, its haecceity has extension $\{a\}$ at o , $\{a_0\}$ at s , and $\{\pi_n a\}$ at each n . Let X^* be the actual LUB of these haecceities. The root Boolean join is coordinatewise in the o, s, n extensions; its u -extension is then determined by the ultrafilter. This gives

$$X_o^* = A_0, \quad X_s^* = \{a_0\}, \quad X_n^* = X_u^* = \{0, 1\}.$$

Take $z = q$ as a proposition-input object, and take a root property Y with $Y_o = \emptyset$, $Y_s = \{a_0\}$, and $Y_n = Y_u = \{0\}$. At s , every X^* -instance necessarily satisfies Y , while $X^*(z) \& \sim Y(z)$ has s/u -profile $(0, 1)$. Thus

$$w \leq [\forall x (X^*(x) \rightarrow \Box Y(x)) \& \text{Atom}(q) \& q \leq (X^*(z) \& \sim Y(z))].$$

These are the preceding witness conditions, with q playing the role of the footnote's w' . Its equality-fiber meet is still bottom.

8. A citation. Footnote 42(ii) should cite Proposition 2.7 rather than 2.6, which presupposes C5. This does not supply the meet-preservation step at issue.

So the issue seems to be the assumed preservation of this meet, rather than a missing necessity operator on Boolean Completeness. The distinction is between existence of internal meets and their preservation under the action maps. The construction has the former, even at every world, but not the latter.